



LM386 AUDIO AMPLIFIER

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Here is a simple audio amplifier circuit built around 8-pin integrated circuit LM386. The author's prototype is shown in Fig. 1.

Circuit and working

Circuit diagram of the LM386 audio amplifier is shown in Fig. 2. It is built around popular amplifier LM386 (IC1), an 8-ohm, one-watt speaker (LS1), four capacitors and a few other components. A 6V battery is used to power this project.

Four electrolytic capacitors [two 10µF, 16V (C1 and C2) and two 220µF, 16V (C3 and C4)] are used

in this circuit. C1 is connected to the middle terminal of 10k potmeter VR1. C2 is connected to pins 1 and 8 of IC1. Pin 5 of IC1 is its output terminal, which is connected to speaker LS1 through C3.

C4 is connected to the positive terminal of 6V battery and ground. Positive side of 6V is connected to pin 6 of IC1 and the other side to ground terminal to pin 4.

Inverting pin 2 of IC1 is connected to ground and non-inverting pin 3 is connected to the input terminal through VR1. Audio input is fed to CON1. VR1 is used to control volume.

Construction and testing

An actual-size, single-side PCB for LM386 amplifier is shown in Fig. 3 and its component layout in Fig. 4. After assembling the circuit on a PCB, enclose it in a suitable box. Fix connector CON1 on the front panel for input and loudspeaker LS1 at the rear side of the box. Connect VR1 on the front panel for controlling the volume.

Before using this project, test it using the 6V battery. Connect the 8-ohm, one-watt speaker to output pin 5 of IC1 through C3. Switch on S1 and keep VR1 to its mid position. Now, take a screw-driver and gently touch it on input terminal pin 3 of IC1. You should hear

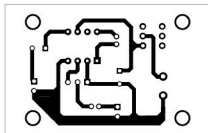


Fig. 3: Actual-size PCB pattern of the audio amplifier

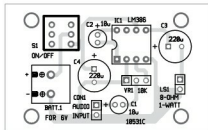


Fig. 4: Component layout of the PCB

PARTS LIST

Semiconductors:	
IC1	- LM386 low-power amplifier
Resistors (all 1/4-watt, ±5% carbon):	
VR1	- 10-kilo-ohm potentiometer
Capacitors:	
C1, C2	- 10µF, 16V electrolytic
C3, C4	- 220µF, 16V electrolytic
Miscellaneous:	
CON1	- 2-pin connector
LS1	- 8-ohm, one-watt loudspeaker
S1	- On/off switch
Batt.1	- 6V battery
	- 2-pin terminal connector for battery

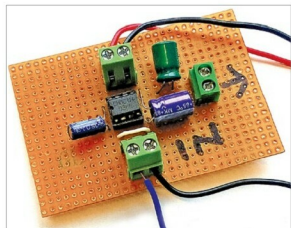


Fig. 1: Author's prototype

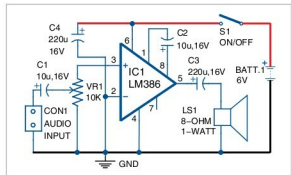


Fig. 2: Circuit diagram of the LM386 audio amplifier

a humming sound from the speaker. This will confirm that your circuit is working and ready to use.

Note. LM386 provides output of 250 milliwatts to one watt depending on supply voltage and load. Refer its data sheet for details. **EFY**



Raj K. Gorkhali is a regular contributor to EFY. His interests include designing electronics circuits